

Intro to Machine Learning



Agenda

- What is Machine Learning ?
- Why use Machine Learning ?
- Types of Machine Learning.
- Main Challenges of Machine Learning.
- Testing and Validation.



What is Machine Learning?



What is Machine Learning?

“Field of study that gives computers the ability to learn without being explicitly programmed”



Arthur Samuel

1959



What is Machine Learning?

“A computer program is said to learn from experience (E) with respect to some task (T) and some performance measure (P), if its performance improves with experience”



Tom Mitchell

1997



What is Machine Learning?

- Machine Learning is the science (and art) of programming computers so they can *learn from data*.





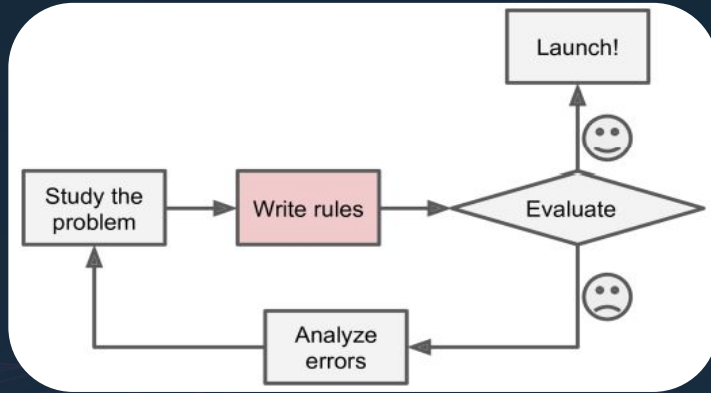
Example : Spam Filter

- If we have examples of spam emails and regular (non-spam) emails (e.g., flagged by users).
- These examples, called the training set, can be fed to a model to learn.
- In this case, the task T is to flag spam for new emails, the experience E is the training data.
- The performance measure P would be the ratio of correctly classified emails. This is called accuracy.



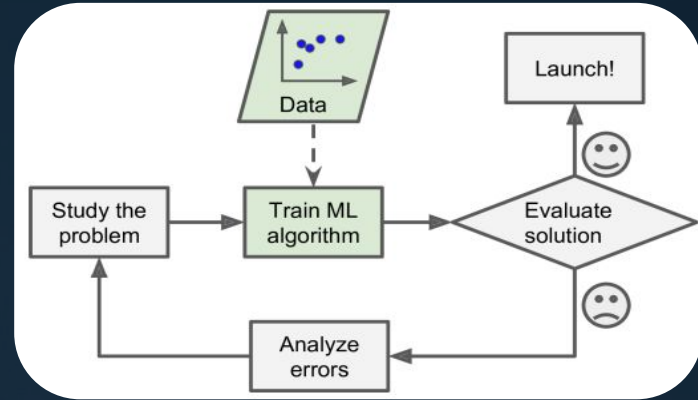
Why use Machine Learning?

Spam Filter Using Traditional Programming



- Rules may include some words (such as “4U”, “credit card”, “free” and “amazing”). Other patterns in the sender’s name, the email’s body, and so on.
- Your program will likely become a long list of complex rules and pretty hard to maintain.

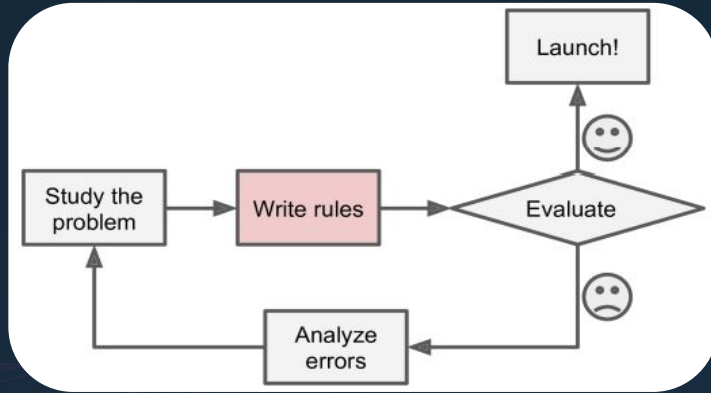
Spam Filter Using Machine Learning



- ML techniques automatically learn which words (or patterns) in the spam examples compared to the ham examples.
- The program is much shorter, easier to maintain, and most likely more accurate.

Images credit: Hands on ML Book

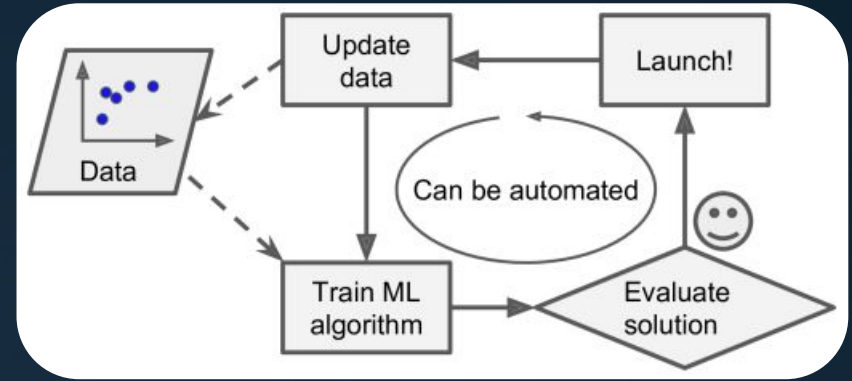
Spam Filter Using Traditional Programming



- If spammers notice that all their emails containing “4U” are blocked, they might start writing “For U” instead. Your Program would need to be updated to flag “For U” emails.
- If spammers keep working around your spam filter, you will need to keep writing new rules forever.

Images credit: Hands on ML Book

Spam Filter Using Machine Learning

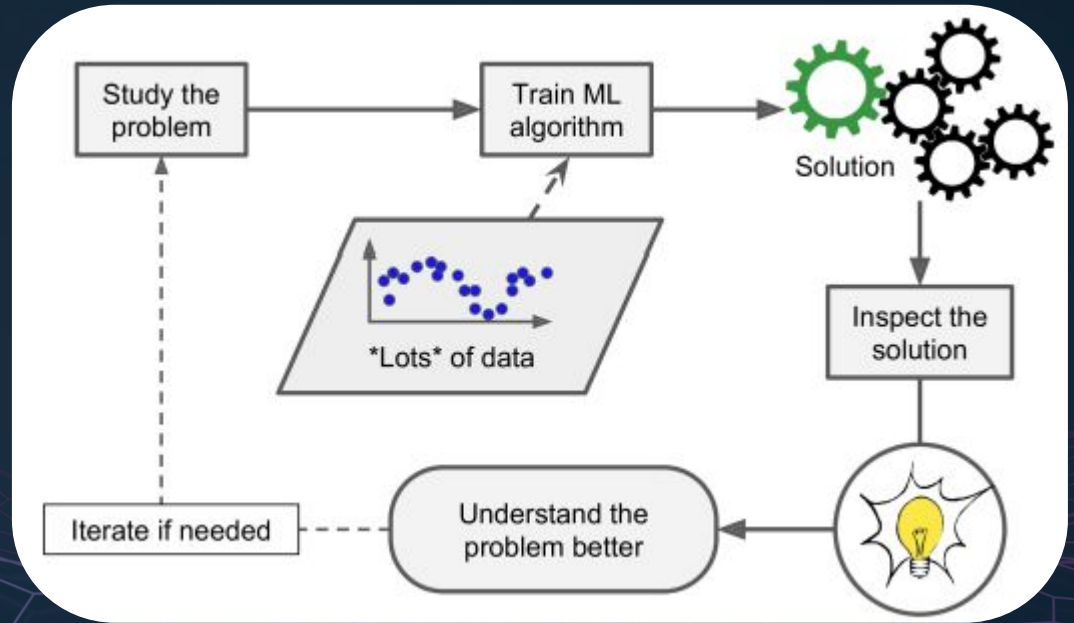


- A spam filter based on Machine Learning techniques automatically notices that “For U” has become unusually frequent in spam flagged by users, and it starts flagging them without your intervention.

Machine Learning is great for:

- Problems for which existing solutions require a lot of hand-tuning or long lists of rules: one Machine Learning algorithm can often *simplify code and perform better*.
- Complex problems for which there is no good solution at all using a traditional approach (Speech Recognition, Self driving vehicles,.. etc.) the best Machine Learning techniques can find a solution.
- Fluctuating environments: a Machine Learning system can adapt to new data.
- Getting insights about complex problems and large amounts of data.

Getting insights about complex problems and large amounts of data.

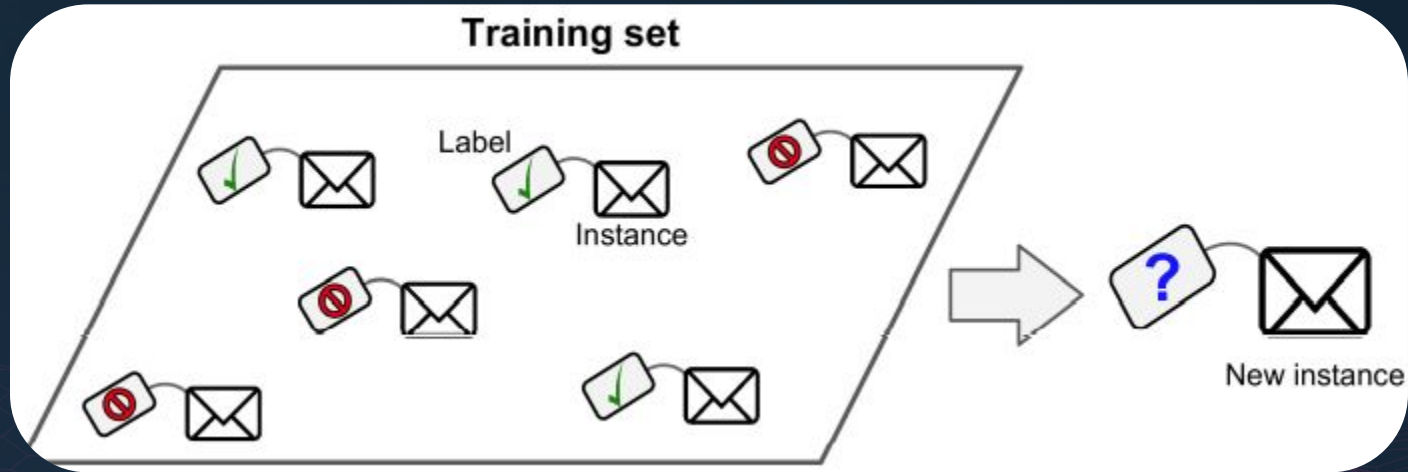


Images credit: Hands on ML Book



Types of Machine Learning

Supervised Learning



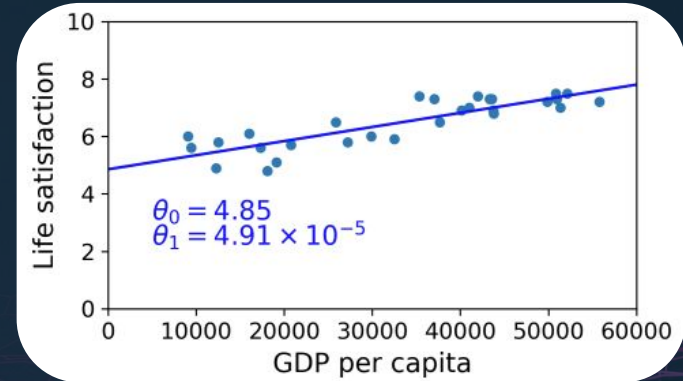
- The training data you feed to the algorithm includes the desired solutions, called labels.
- When Labels are categories (classes), it is called ***“Classification”***.

Images credit: Hands on ML Book

Supervised Learning

- When Labels are numeric values (Continuous), it is called “**Regression**”

Country	GDP per capita (USD)	Life satisfaction
Hungary	12,240	4.9
Korea	27,195	5.8
France	37,675	6.5
Australia	50,962	7.3
United States	55,805	7.2

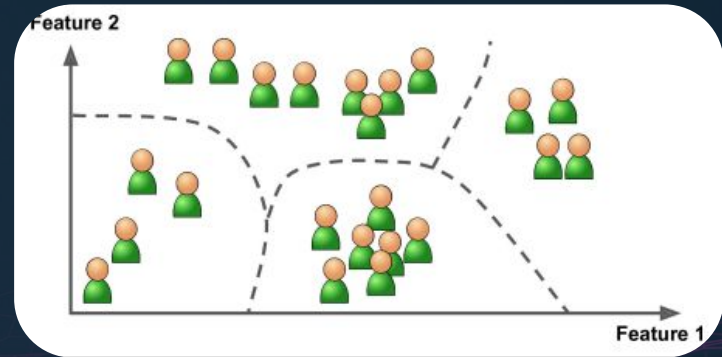
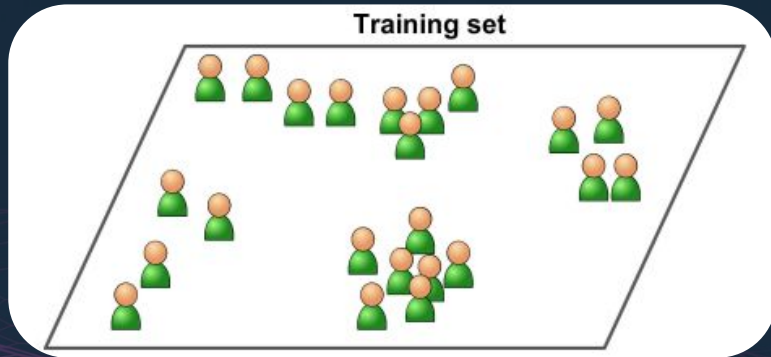


$$\text{life_satisfaction} = \theta_0 + \theta_1 \times \text{GDP_per_capita}$$

Images credit: Hands on ML Book

Unsupervised Learning

- The training data is unlabeled.
- The system tries to learn *without a teacher*.



Clustering

- Say you have a lot of data about your blog's visitors. You may want to detect groups of similar visitors

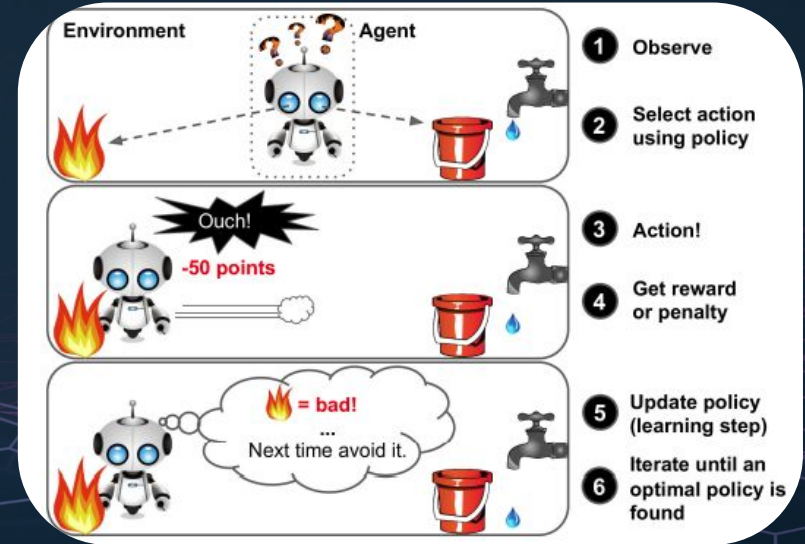
Images credit: Hands on ML Book

Semi Supervised Learning

- When dealing with *partially labeled training data*, usually a lot of unlabeled data and a little bit of labeled data.

Reinforcement Learning

- The learning system, **agent**, can observe the environment, select and perform actions, and get rewards in return (or penalties).
- It learn by itself what is the best strategy, called a policy, to get the most reward over time.



Images credit: Hands on ML Book



Main Challenges of Machine Learning



Main Challenges of Machine Learning

- Insufficient Quantity of Training Data





Main Challenges of Machine Learning

- Nonrepresentative Training Data (Distribution)





Main Challenges of Machine Learning

- **Poor-Quality Data**

(Errors, Missing Values, Noise, Outliers,...etc)

→ Data Cleaning

→ Data Preprocessing

- **Irrelevant Features**

garbage in, garbage out

→ Feature Engineering (Feature Extraction)

→ Feature Selection



Main Challenges of Machine Learning

- **Overfitting the Training Data**
(Too complex model)
 - Simplify the model
 - Collect More Data
- **Underfitting the Training Data**
(Too simple model)
 - More Complex model
 - Better Data (feature Engineering)



Main Challenges of Machine Learning

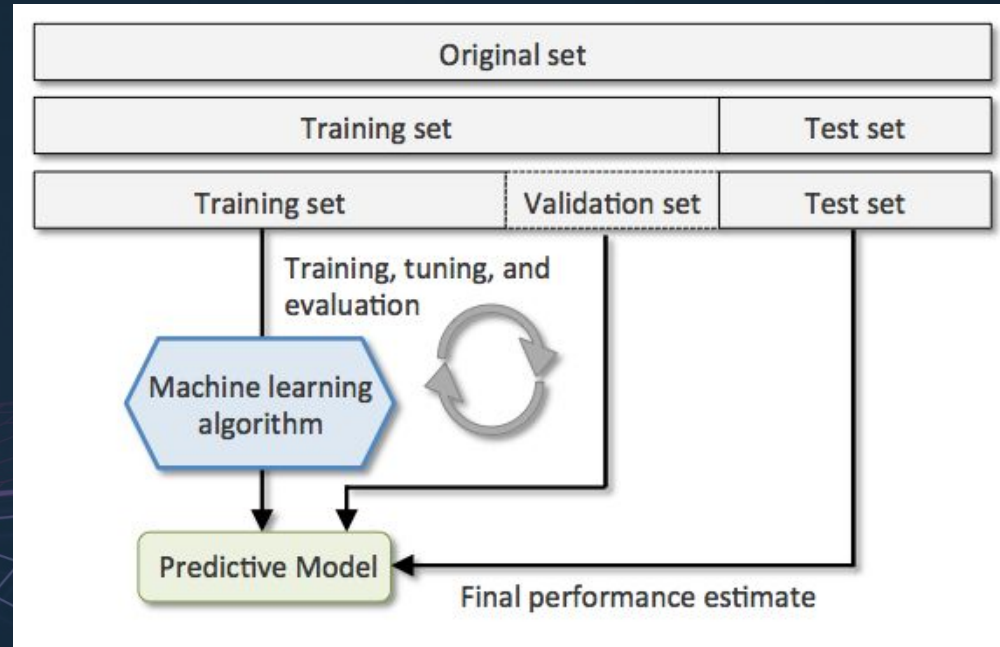
- Insufficient Quantity of Training Data
- Nonrepresentative Training Data
- Poor-Quality Data
- Irrelevant Features
- Overfitting the Training Data
- Underfitting the Training Data



Testing & Validation



Testing and Validation



Images credit: ML Bookcamp

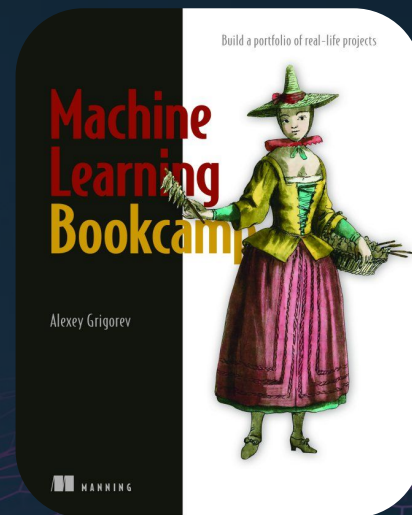
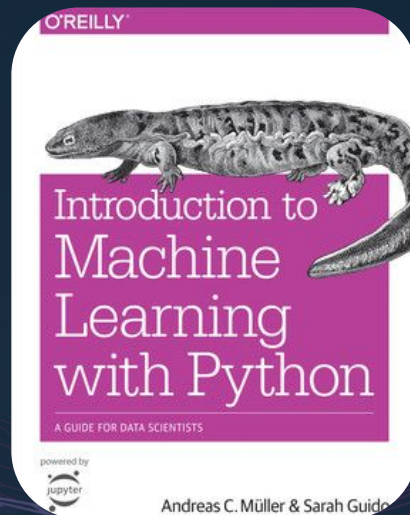
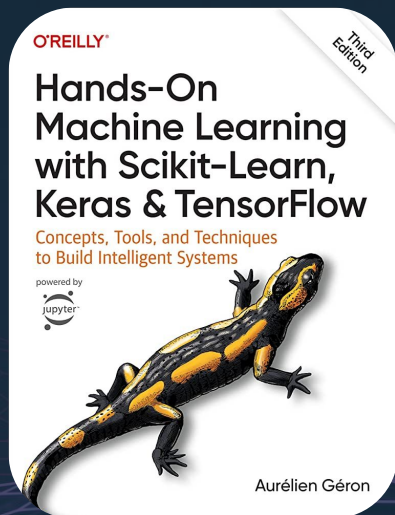


References



References

Main Reference





Happy Learning!

